

Southeast University Examination Paper (MidTerm Self-test)

Course Name Principles of Compiling Examination Term 16-17-2 Score _____Related Major _____ Examination Form Close test Test Duration 120 Mins

There are 5 problems in this paper. You can write the answers in English or Chinese.

1. Please construct **context-free grammars without ϵ -productions** for the following language.(20%)

$L = \{a^m \omega b^m \mid m \geq 0 \text{ and } \omega \in (c,d,e,f)^* \text{ and the numbers of } c\text{'s and } d\text{'s and } e\text{'s occurred in } \omega \text{ are even}\}$

2. Please construct a **DFA with minimum states** for the following regular expression. (20%)

$((a|b)^*(ab))^*(a|b)^*(a|b)$

3. Please **eliminate the left recursions (if there are)** and **extract maximum common left factors (if there are)** from the following context free grammar, and then decide **the resulted grammar** is whether a LL(1) grammar by **constructing the related LL(1) parsing table**.(20%)

$S \rightarrow \text{if } E \text{ then } S \mid \text{if } E \text{ then } S \text{ else } S \mid \text{while } E \text{ do } S \mid a$

$E \rightarrow E \text{ and } E \mid E \text{ or } E \mid \text{not } E \mid (E) \mid b$

4. Please **construct a LR(1) parsing table for the following ambiguous grammar with your own defined additional conditions (You determine the required additional conditions by yourself).**(20%)

$S \rightarrow \text{if } E \text{ then } S \mid \text{if } E \text{ then } S \text{ else } S \mid a$

$E \rightarrow E \text{ and } E \mid \text{not } E \mid (E) \mid b$

5. **Prove** that if G is a LL(1) grammar, then it is a LR(1) grammar.(20%)